

Anshul Raje

arraje@wpi.edu | +1 (508) 450-7630 | LinkedIn | GitHub | Portfolio | Boston, MA

EDUCATION

M.Sc. in Robotics Engineering, Worcester Polytechnic Institute, Worcester, MA

Aug 2026 – May 2028

Courses: Vision-based Robotic Manipulation, Hands-on Autonomous Aerial Robotics

B.E. in Electronics and Communication Engineering, BITS Pilani, Goa, India

Sep 2020 – Jul 2024

GPA: 3.40/4.0 | Courses: Digital Image Processing, Control Systems, Digital Signal Processing

SKILLS

Languages: Python, C/C++

Frameworks & Libraries: ROS 2, ROS, ROS 2 Control, OpenCV, Ultralytics

Tools & Platforms: Linux, Git, Docker, Gazebo (robot simulation), RViz (visualization)

Hardware: NVIDIA Jetson (Orin/Xavier/Nano), Depth Cameras (Stereolabs ZED, Intel RealSense)

EXPERIENCE

Strider Robotics | Associate Robotics Engineer | Bangalore, India

Aug 2024 – Apr 2026

- Automated analog gauge reading by detecting needle base, needle tip, and the min and max scale markers, then converting the angle between them into a calibrated value, reading 22 of 25 gauges correctly per run on site.
- Built a vision-guided capture pipeline servoing a PTZ camera onto a known ArUco marker and then onto the gauge itself, zooming and autofocusing to image each asset from a repeatable viewpoint every round.
- Cut inference cost in the alignment loop by running YOLO with DeepSORT tracking only on the first pass, then switching to normalized cross-correlation template matching for every later alignment step.
- Trained YOLOv12 detectors on 2500+ hand-labelled images spanning gauge, needle, scale marker, and fire extinguisher classes, converting them to TensorRT engines to run 720p inference at 25 FPS onboard a Jetson.
- Designed a five-layer ROS 2 perception architecture separating camera drivers, computer vision inference nodes, and task orchestration, and deployed it on legged inspection robots at two industrial clients.
- Architected a behavior tree orchestration framework separating reusable perception primitives from client-specific task sequences, letting the team add a new client's inspection routine as a configuration package in one to two hours.
- Tools & Techniques:** Visual Servoing, ArUco Marker Detection, Multi-Object Tracking, Template Matching, TensorRT, Edge Inference, Behavior Trees, DeepSORT.

HiPeRT Lab, UNIMORE | Robotics Research Intern | Modena, Italy

Jul 2023 – Dec 2023

- Engineered a synthetic dataset pipeline segmenting track footage with Segment Anything Model (SAM) and compositing opponent car CAD models per frame, auto-annotating 15000+ images with zero manual labeling.
- Trained YOLOv4 on domain-randomized frames varying lighting, saturation, and blur, hitting a 97% detection rate on live on-track video and adding camera perception to a LiDAR-only race car stack.
- Tools & Techniques:** Synthetic Data Generation, Domain Randomization, Real-Time Inference, SAM, YOLOv4.

PROJECTS

Attitude Estimation using a 6-DoF IMU [\[Report\]](#) [\[Video\]](#)

Aug 2026 – Sep 2026

- Benchmarked five attitude estimation methods against Vicon motion capture ground truth across six flight sequences, with an unscented Kalman filter reaching the lowest mean orientation RMSE of 14.6 degrees.
- Implemented a quaternion-based unscented Kalman filter fusing accelerometer data with a bias-calibrated gyroscope, propagating sigma points through the nonlinear process model to hold roll and pitch error to 5.4 and 4.8 degrees.
- Tools & Techniques:** State Estimation, Sensor Fusion, Unscented Kalman Filter (UKF), Madgwick Filter, Complementary Filter, Quaternion Attitude Representation, IMU Calibration, RMSE Benchmarking.

Autonomous Navigation and Manipulation for a Mars Rover [\[Code\]](#) [\[Video\]](#)

Sep 2021 – May 2023

- Added real-time obstacle detection to an autonomous rover by training YOLOv3 on a custom dataset, letting the rover identify and avoid hazards on course without operator input.
- Implemented a Stanley controller for autonomous waypoint navigation, holding tracking error within 5 cm across the traversal course.
- Developed forward and inverse kinematics for a 5-DOF onboard manipulator, giving operators end-effector position control instead of commanding each joint separately.
- Tools & Techniques:** Object Detection, Real-Time Inference, Waypoint Navigation, Stanley Controller, Forward & Inverse Kinematics, YOLOv3.

ACHIEVEMENTS

- Placed 2nd globally at the Anatolian Rover Challenge and won Best Autonomous and Controls (Jul 2022).
- Won the Excellence Award for Overall Performance at the International Rover Challenge, Bangalore (Jan 2023).